Justin Jasper

(770) 480-8093 | jtjasper@stanford.edu |https://www.linkedin.com/in/justin-jasper

EDUCATION

Stanford University Stanford, CA
Master of Science in Computer Science (Concentration: Artificial Intelligence) *June 2026*

Stanford University Stanford, CA
Bachelor of Science in Bioengineering *June 2025*

EXPERIENCE

Centre for Artificial Intelligence and Robotics Hong Kong, China
Biomedical AI Intern *June 2025 - September 2025*

- Built full-stack AI tools to improve the accuracy of echocardiogram diagnoses and to support clinical decision-making + treatment plan formulation (Pytorch, Python, React, PostgreSQL, Neo4j)

MITRE Corporation Mclean, VA
Computational Biology Intern *June 2024 - December 2024*

- Built agentic AI tools to enable extensive scaling of synthetic biology experiments and industrial workflows.

Omniwear AI Palo Alto, CA
CoFounder, CEO *December 2022 - December 2023*

- Led the development of a virtual try-on platform for e-commerce powered by computer vision and generative AI. Participated in Launchpad (Stanford's premier graduate accelerator) Spring '23 and Entrepreneur First's NYC unicorn training program. Piloted the tech with Abercrombie & Fitch, Revolve, and Everlane.

PROJECTS

CV Image Processing for CT and Ultrasound-based Tumor Detection

- Built a Python/OpenCV pipeline to segment and reconstruct 3D lung CT scans for cancer diagnosis, training an Attention U-Net (TensorFlow) for malignant/benign classification and applying K-means clustering to segment pulmonary abnormalities.

Python-Based Probabilistic Modeling of Biological Systems

- Built Python computational models simulating molecular systems — including ion channel activation, protein-ligand binding, and polymer conformations — using Boltzmann statistical mechanics to predict multi-state behaviors and system energetics.

Computational Analysis of NMDA Receptor Antagonist Binding for Chronic Pain Therapeutics

- Used SwissDock and AutoDock Vina to model NMDA receptor antagonist binding interactions, optimizing docking workflows and quantifying binding affinities to evaluate drug repurposing candidates for chronic pain.

Synthetic Biosensor Development for Pathogen Detection

- Engineered a novel Salmonella biosensor using a SynNotch receptor system with an anti-Salmonella scFv transfected into HEK293 cells, validated via flow cytometry, Western blotting, PCR, and gel electrophoresis.

Computer Vision for Meal Nutritional Analysis

- Built a full-stack web app (Flask + HTML) for real-time food nutritional analysis from user-uploaded images, powered by a ResNet18 CNN with an object detection pipeline for food localization and classification.
- https://github.com/justinjasper/cv-nutritional-analysis

SKILLS

- Proficient in Python (NumPy, Pandas, SciKit-learn, TensorFlow), C++, C, MATLAB, git, UNIX, LLM training & deployment (LangChain, LlamaIndex, Kubernetes), retrieval augmented generation (RAG), neural networks
- Proficient in Spanish, with strong cross-cultural communication skills.